

pumping recovery analogue summary

Pumping response figures (analogues of drought recovery)

Date: August 2026

Script: redo_pumping_recovery_analogues.py

Deck: recovery_and_pumping_figures.pptx (drought + pumping; open in Google Slides via File -> Import)

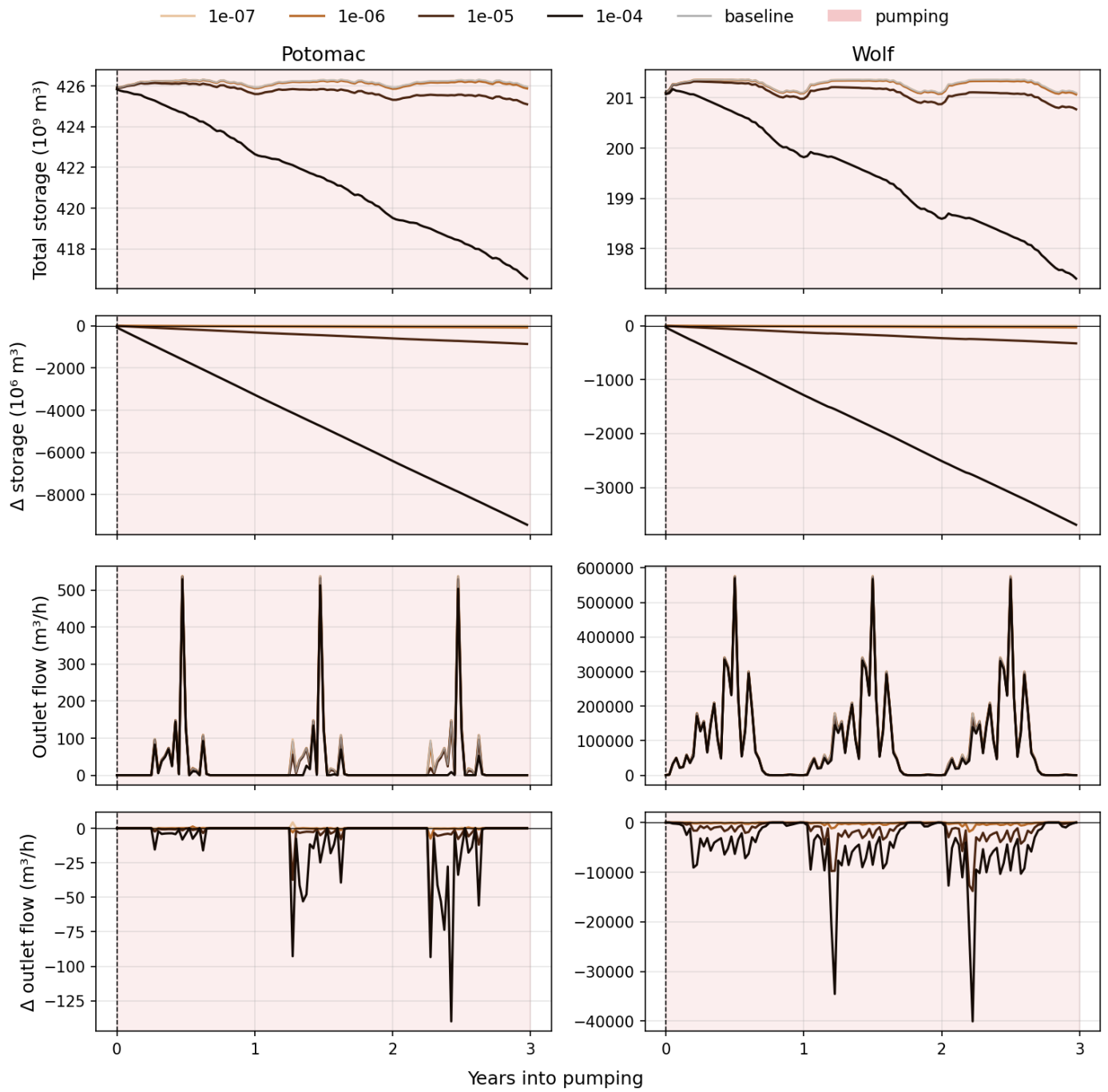
Ensemble design adjustments

3_year_pumping_tests differs from droughts:

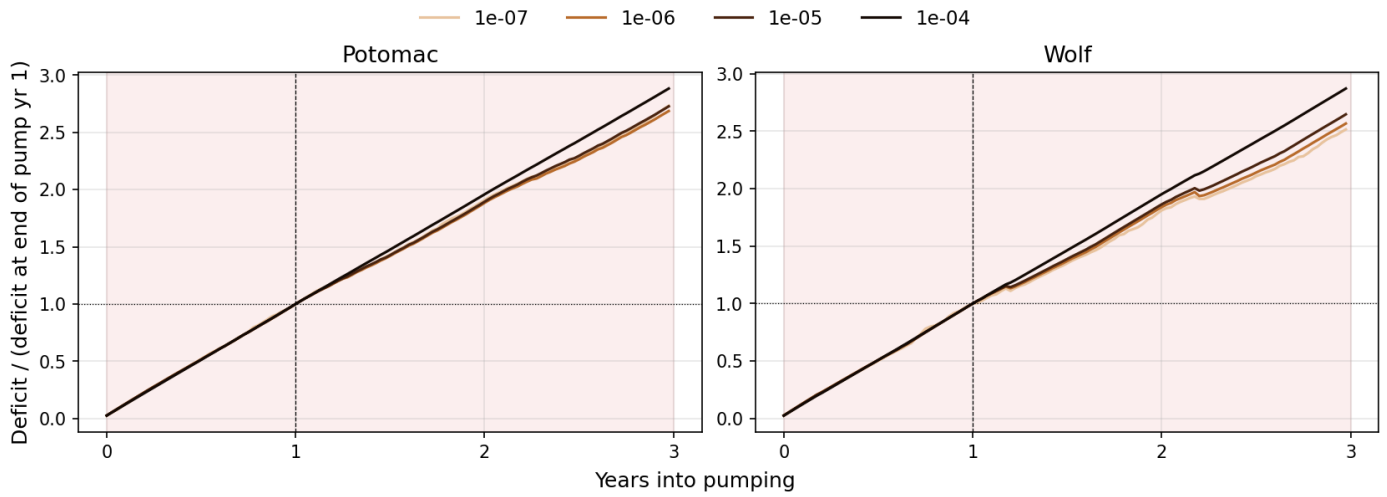
	Droughts	Pumping
--	-----	-----
Stress axis	Duration (1 / 3 / 10 / 50 yr)	Rate (1e-7 ... 1e-4 m/h domain-avg)
Sequence	40 spinup -> drought -> 5 yr recovery	40 spinup -> 3 yr pumping (ends while still pumping)
Temp / persist	Recovered in recovery yr 1 vs remaining	Early (end pump yr 1) vs buildup (extra by end yr 3)
Baseline	short_baseline / baseline	droughts/short_baseline (shared spinup hashes)

Figures

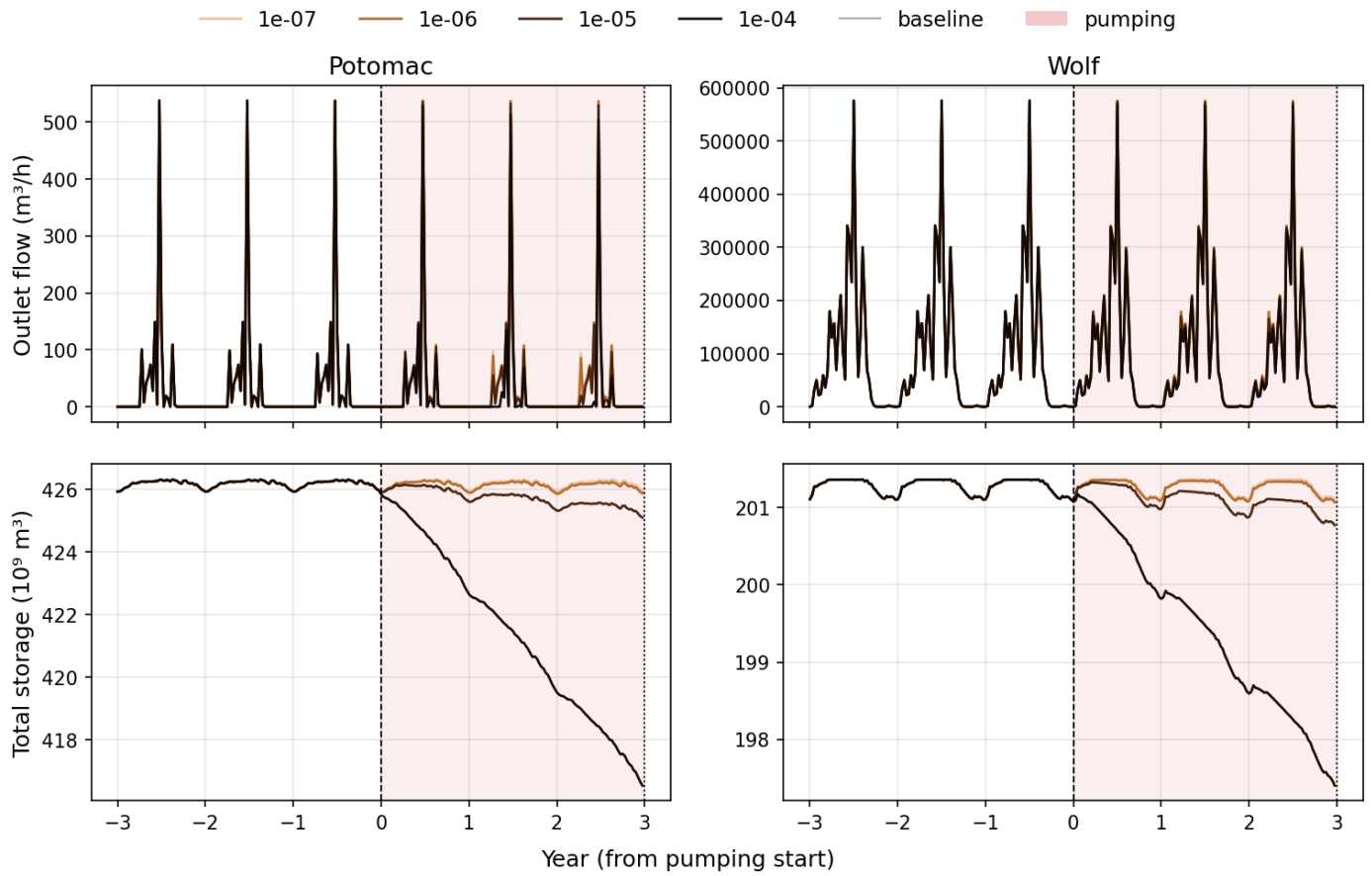
- pumping_during_totals_and_anomalies.png



- pumping_by_rate_fractional_storage.png



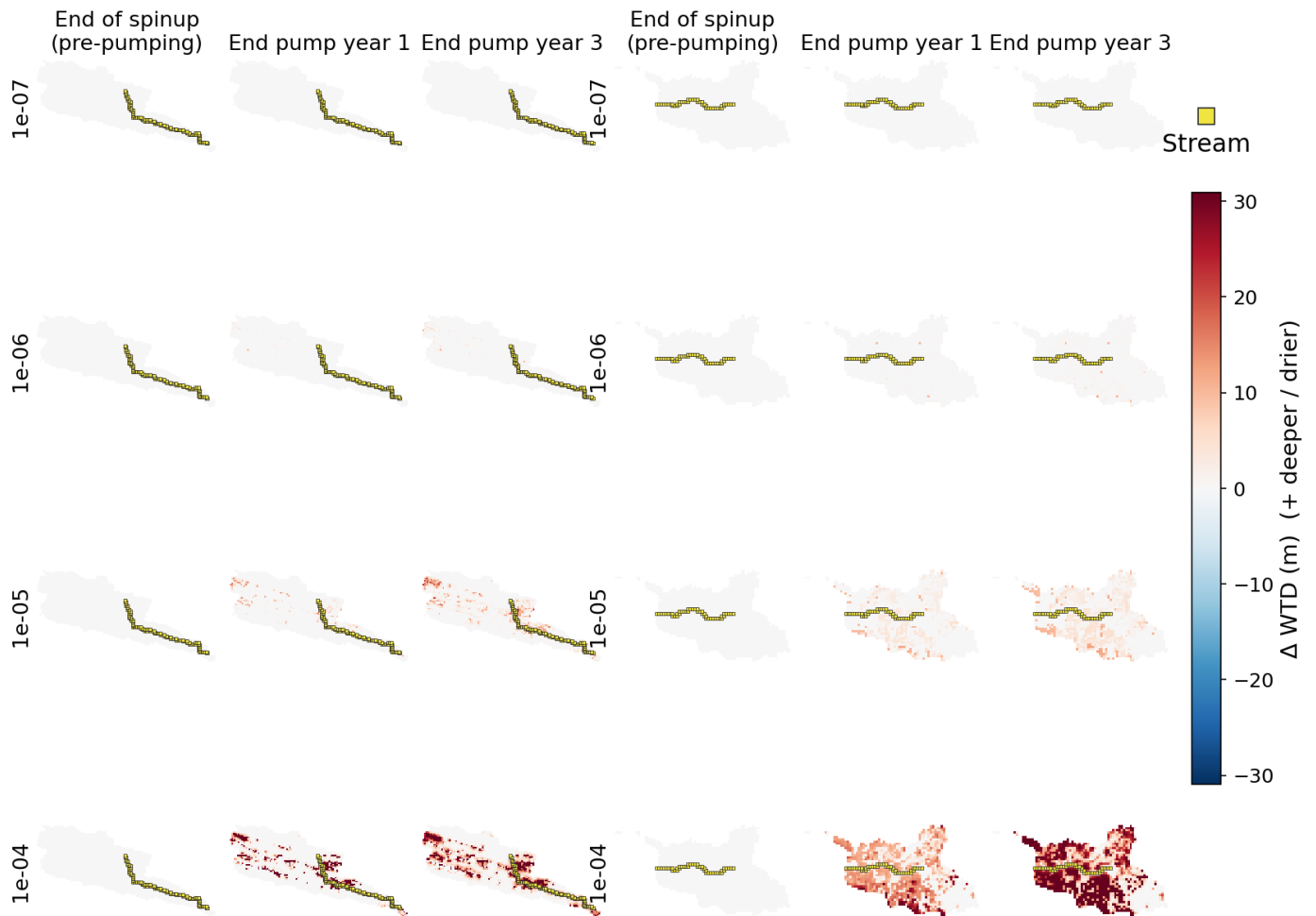
- pumping_course_streamflow_storage.png



- pumping_wtd_anomaly_maps.png

Potomac

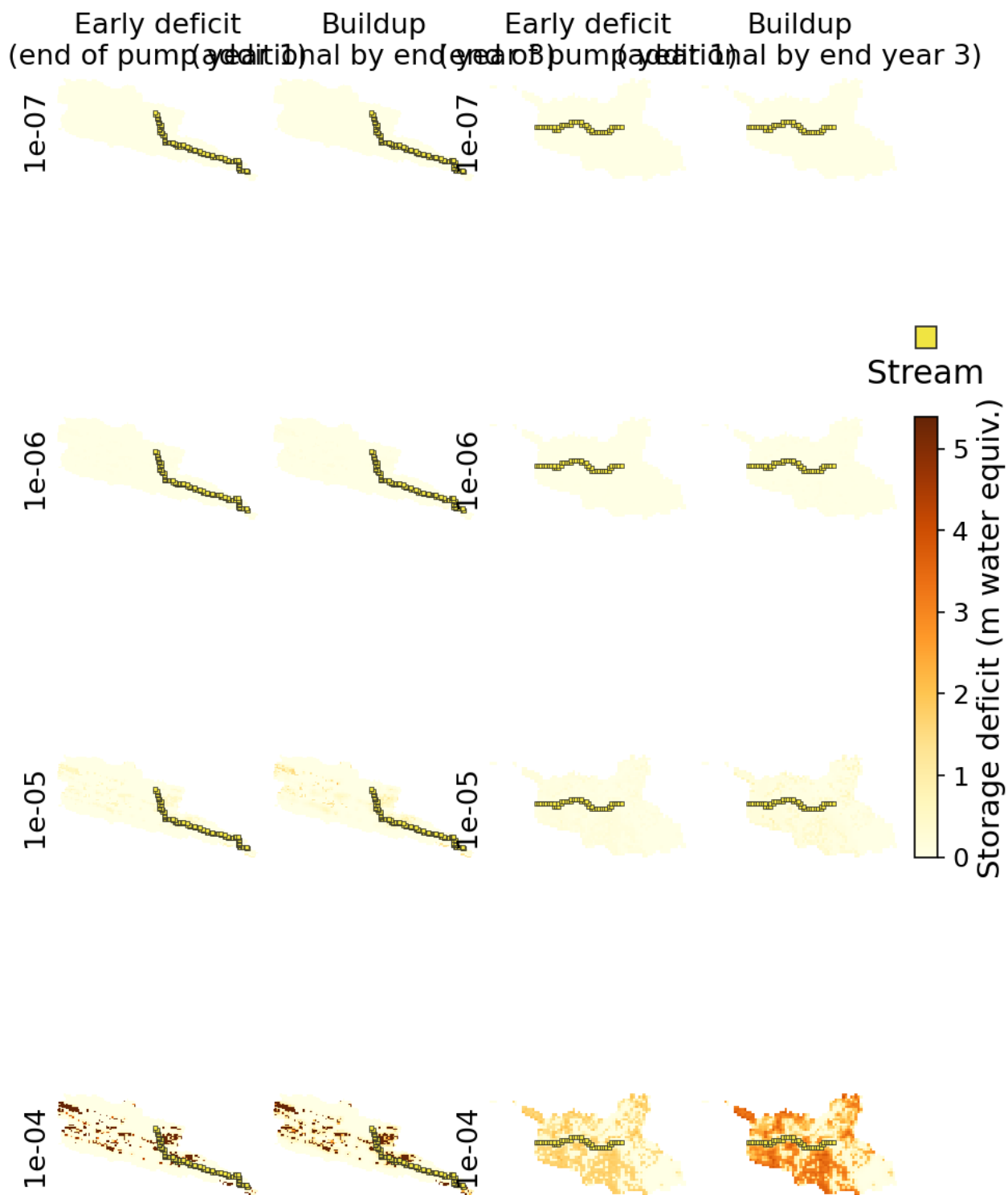
Wolf



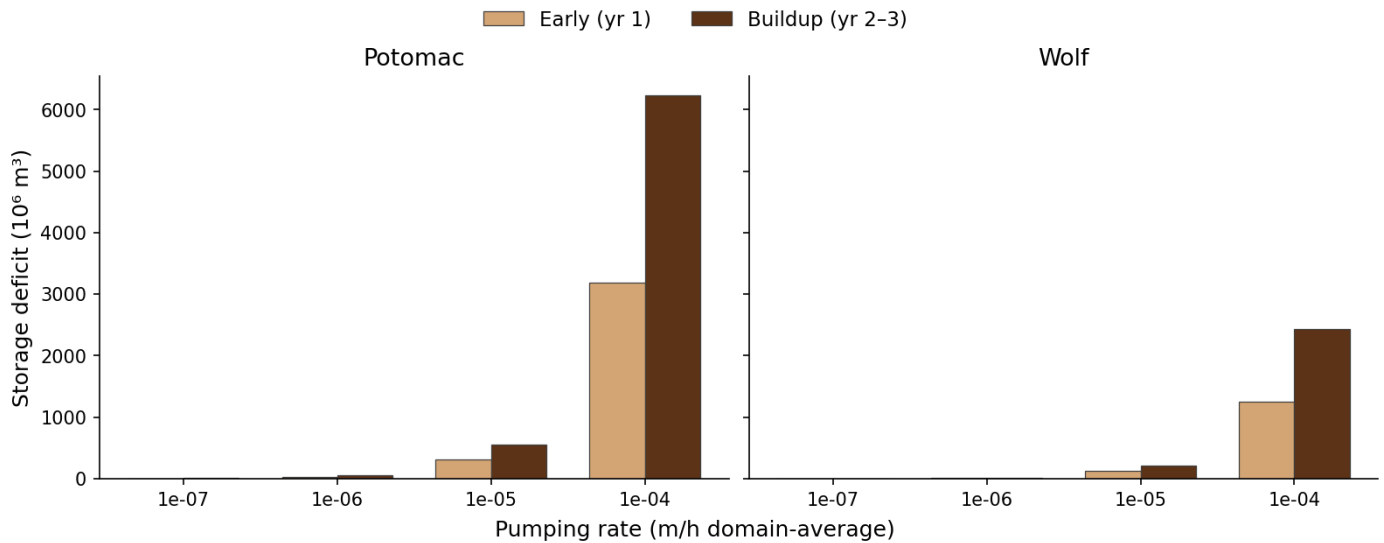
- pumping_early_buildup_storage_deficit_maps.png

Potomac

Wolf



- pumping_early_buildup_deficit_bars.png



Domain totals (106 m3) - early / buildup

Rate	Potomac	Wolf
1e-7	3.5 / 5.9	1.3 / 2.1
1e-6	31 / 54	12 / 20
1e-5	308 / 554	120 / 206
1e-4	3188 / 6235	1254 / 2437

Deficits scale roughly linearly with rate; buildup over years 2-3 exceeds the year-1 early deficit at all rates.

Reproduce

```

```bash
conda activate /glade/work/bwest/conda-envs/droughts
source ~/pf_env.sh
cd /glade/derecho/scratch/bwest/drought-ensemble
python analysis/redo_pumping_recovery_analogues.py
python analysis/build_recovery_pumping_pptx.py
```

```